Prompt 3: **The final version of the evaluation prompt.** It is enriched with the few-shot prompting technique and imposes a logical compliance with provided labels.

```
You are a logic module designed to provide accurate answers.
In a Bongard Problem the objective is to spot the difference between the contents of
↪  images located on the two opposite sides of the problem.
You are given correct labels of these sides and must decide whether the answer provided
↪  by the user is correct and matches with those labels. Answer with 'OK' or 'WRONG'.
The user's answer has to strictly logically match the labels, as shown in examples.

FIRST EXAMPLE:
LEFT SIDE LABEL: All shapes are small.
RIGHT SIDE LABEL: All shapes are big.
USER ANSWER: On the left side, one of the shapes is small. On the right side, all of the
↪  shapes are big.
EVALUATION: WRONG
END OF FIRST EXAMPLE.

SECOND EXAMPLE:
LEFT SIDE LABEL: All shapes are small.
RIGHT SIDE LABEL: All shapes are big.
USER ANSWER: On the left side, all of the shapes are small. On the right side, all of
↪  the shapes are big.
EVALUATION: OK
END OF SECOND EXAMPLE.

LEFT SIDE LABEL:
<left_label>

RIGHT SIDE LABEL:
<right_label>

USER ANSWER:
<model_answer>
```

Table 5: **Consensus with manual evaluation on synthetic BPs.** The percentage of the solutions evaluated the same as in our manual evaluation in BP instances $\#1 - \#100$ (Bongard, 1970). The assessed solutions were obtained by GPT-4o using the *Descriptive* prompting strategy.

INITIAL PROMPT			FINAL PROMPT		
ALL MODELS	ANY MODEL	VOTING	ALL MODELS	ANY MODEL	VOTING
0.93	0.90	0.94	0.90	0.96	0.99

refers to the cases where a solution is marked as correct if at least one model evaluates it as correct. *Voting* refers to the hard-voting scheme described in section E.1. It is important to observe that the chosen voting evaluation method differed from the manual evaluation in only one specific problem, which is depicted in Fig. 12.

In addition, we checked the performance of our enhanced evaluation prompt on 20 new, not used in other experiments, manually evaluated Bongard-OpenWorld problems solved by GPT-4o using the *Descriptive* strategy. Again, the use of Prompt 3 visibly increased the consensus with manual evaluation (see Table 6). The difference between our manual evaluation and the voting scheme occurred only in 2 problem solutions whose correctness is disputable (see Fig. 13).

While the choice of examples shown in the prompt may in some cases impact the evaluation performance, the finally proposed evaluation prompt seems to be well suited to both domains: synthetic and real-world, and should potentially be effective in other similar datasets and solving strategies.

F. Bongard-RWR Dataset

The dataset generation algorithm is presented in Algorithm 1 using notation introduced in Section 4.

Furthermore, Fig. 15 provides additional examples of the proposed Bongard-RWR dataset. Each subfigure presents a comparison between the synthetic Bongard problem and its respective real-world translation in Bongard-RWR. Examples

Table 6: **Consensus with manual evaluation on Bongard-OpenWorld.** The percentage of the solutions evaluated the same as in our manual evaluation in the additional 20 Bongard-OpenWorld instances $\#101 - \#120$ (Wu et al., 2024), not used in the main experiment. The solutions were obtained by GPT-4o using the *Descriptive* prompting strategy.

INITIAL PROMPT			FINAL PROMPT		
ALL MODELS	ANY MODEL	VOTING	ALL MODELS	ANY MODEL	VOTING
0.75	0.70	0.70	0.65	0.85	0.90

15a and 15b were translated automatically, whereas 15c and 15d were constructed fully manually, including building an appropriate scene and taking a picture. Additionally, Fig. 14 shows a particular approach taken when translating a given synthetic BP to its Bongard-RWR counterpart (problems not translated and those rejected after translation are combined into one category).

G. Coverage of Bongard-RWR Instances

Even though the final results of individual models and strategies solving Bongard-RWR are somewhat unsatisfactory, especially in the case of open language response generation, it is worth to examine the degree of overlap of correct answers across the tested models. On the one hand, the results of ground-truth classification and model's disagreement on the solution evaluation clearly confirm inability of any single model to solving all problems from the Bongard-RWR dataset. On the other hand, it is likely that the overlap is not complete, and it is possible to expand the solution coverage by appropriate mixture-of-experts approach. This is indeed the case in our experiments, as illustrated in Figs. 16 – 23. While single proprietary MLLMs solved between 9 and 14 instances, the number of instances solved by *any* MLLM equaled 23. We leave exploration of this path for future research.

H. Comparison of Synthetic Bongard vs. Bongard-RWR results

Our research shows that all of the tested models have difficulty solving synthetic concepts when applied to real-world images. Comparing the results for both datasets (see Figs. 17 and 23) we identified some discrepancies. Four problems that remained unsolved in the synthetic BPs were successfully solved in the real-world domain of the Bongard-RWR dataset: #56, #87, #88, and #98. However, three of these problems differ slightly from their synthetic counterparts. The images in #56 from Bongard-RWR feature a variety of colors instead of the usual black-and-white figures. Furthermore, in real-world version of problem #87 more images feature disjoint elements instead of multi-part objects, which may have nudged the model toward the correct answer. Additionally, in problem #98, the figures are shown against a hatched texture, which was not accounted for in the real-world translation.

Conversely, 22 problems that were solved in the synthetic BPs were not solved in the Bongard-RWR dataset. In most cases, the models focused on general associations that did not apply to all the images. For example, the concept presented in problem #3 is: "LEFT: Outline figures, RIGHT: Solid figures." Claude 3.5 Sonnet, using the Descriptive strategy, responded: "LEFT: focuses on practical, everyday objects or scenes, RIGHT: emphasizes aesthetic, artistic, or decorative elements that serve more for visual appeal than utility". Nevertheless, both sides feature a red coffee cup with a saucer, which matches both generated descriptions. The key difference lies in the color of the saucer's rim, which determines whether it should be considered as an outline or a solid figure.

I. MLLM Prompts

I.1. Prompts for Bongard-RWR Generation

At first, Prompt 4 was employed to generate real-world representations of synthetic concepts. After downloading the images corresponding to these textual descriptions, Prompt 5 was used to select those images that correctly represent given concept translation. In addition to the left and right concepts, we also provided prompts briefly explaining the context that the image should match. These prompts were generated during the translation stage of our algorithm (see the fourth line in Algorithm 1).

Algorithm 1 The Bongard-RWR dataset generation.

Input: A set of synthetic concepts Y
Output: A set of generated instances $\mathcal{RWR}$
$M \leftarrow 15, \quad N \leftarrow 10, \quad T \leftarrow 3$
for $y^X \in Y$ **do**
  $D^X \leftarrow$ GenerateTranslations(y^X, N)
  $I^X \leftarrow \emptyset, \quad P \leftarrow \emptyset$
  for $D_i^X \in D^X$ **do**
    for $m \leftarrow 1$ to M **do**
      for $S \in \{L, R\}$ **do**
        $I \leftarrow$ DownloadImage(D_i^{XS}, m)
        if I is accepted by model **then**
          $I_i^{XS} \leftarrow I_i^{XS} \cup \{I\}$
        end if
      end for
      if $|I_i^{XL}| \geq 2$ and $|I_i^{XR}| \geq 2$ **then**
        $P \leftarrow P \cup \{i\}$
        break
      end if
    end for
    if $|P| \geq T$ **then**
      Break
    end if
  end for
  $\mathcal{L}^X \leftarrow \emptyset, \quad \mathcal{R}^X \leftarrow \emptyset$
  if $|P| \geq T$ **then**
    for $k \leftarrow 1$ to 6 **do**
      $p \leftarrow P[k \mod T]$
      $j \leftarrow k \div T$
      for $S \in \{L, R\}$ **do**
        $\mathcal{S}^X \leftarrow \mathcal{S}^X \cup \{I_p^{XS}(j)\}$
      end for
    end for
    $\mathcal{RWR}^X \leftarrow \{\mathcal{L}^X, \mathcal{R}^X\}$
    $\mathcal{RWR} \leftarrow \mathcal{RWR} \cup \{\mathcal{RWR}^X\}$
  end if
end for

I.2. Prompt Describing the Bongard Problem

Prompt 6 describing the BP task has been placed at the beginning of each solving strategy introduced in Section 3.1.

I.3. Prompts for Classification Strategies

Prompt 8 was used to assess solution correctness (see Section 3.2). Prompt 7 was used to assign images to sides (see Section 3.3).